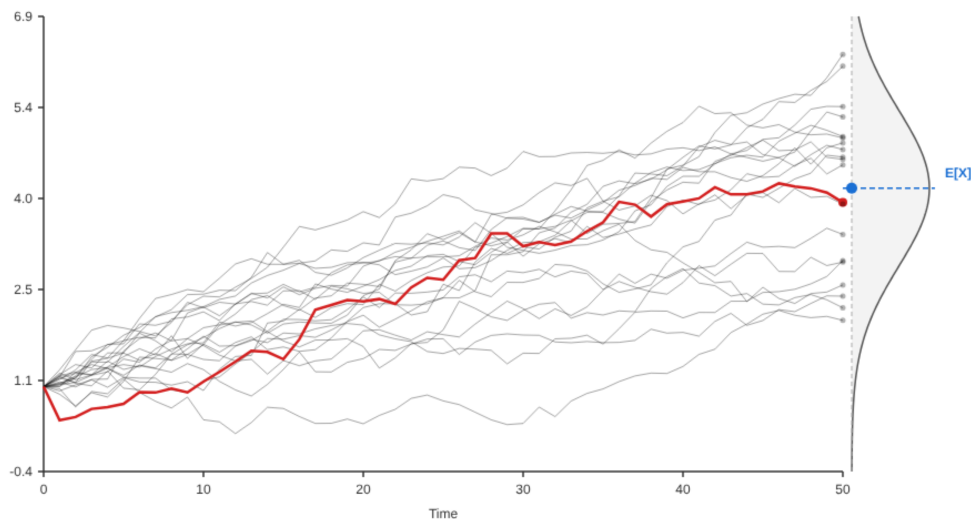


Ergodicity

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Ergodicity is an interesting subtlety to consider when modeling time-varying random processes, like wealth accumulation or whether a gamble is favorable. A stochastic process is ergodic, if its ensemble average across realizations equal the long-run time average along a single trajectory. Let's say we want to assess what is a likely outcome of an ergodic process after t periods from now. We can simulate the situation many times and see what the average result is on period t .



If the process is ergodic, as in the figure above, the mean across runs (blue) is informative of what to expect on an individual run (red). If the process is not ergodic, this approach does not lead to correct information! Instead, it can be the case that you are expected to go down despite the simulation average going up.

Consider a **gamble** modified from Ole Peters (2018): you bet a sum of money and a fair coin is used to either increase it by 50% or reduce it by 40%, with one dollar being the smallest permitted bet. This gamble is used to demonstrate the difference between an ergodic (case 1) and non-ergodic (case 2) processes. First we go through the gamble conceptually, then simulate it with R.

Case 1: Ensemble average

Consider a group of friends who gamble for 100 days in a row, each of them starting with 100\$. Everyone plays the gamble once per day, betting all they have. Every evening they pool their money and redistribute the results evenly among themselves.

How much money can each of the friends expect to have after the 100 days? We can calculate this as the **ensemble average** over the friends. On the first day, each one is expected to have

$$0.5 * 1.5 + 0.5 * 0.6 = 1.05$$

more money than they started with, that is $1.05 * 100 = 105\$$. After 100 days the multiplier becomes

$$1.05^{100} = 131.5$$

yielding an expected wealth per person of

$$131.5 * 100 = 13150\$$$

Each period the mean wealth of each friend in the group is expected to increase by 5%, compounding multiplicatively for 100 days. Hence, the gamble looks very profitable.

Denote wealth by W_{it} , where the number of friends are indexed by $i = 1, \dots, N$ and time by $t = 1, \dots, T$.

Ensemble average at time t is calculated as

$$\bar{W}(t) = \frac{1}{N} \sum_{i=1}^N W_i(t)$$

and the time average is given by

$$\bar{W}(i) = \frac{1}{T} \sum_{t=1}^T W_t(i)$$

Under ergodicity, and in the limit, these two give the same result

$$\bar{W}(t) = \bar{W}(i)$$

Someone's losses are more than covered by the winnings of someone else.

Case 2: Time average

Now consider the same exact gamble and time frame, but instead of a group of friends, one person is gambling alone. Each day she bets all the money she has gained or has left. If she ends up having less than one dollar left, she can't continue gambling.

How much money is she expected to have after the 100 days? Surprisingly, she will go bust! This time we have to calculate the **time average**, as there is no ensemble. Let n_1 represent the number of victories in the gamble, yielding the +50 %, and n_2 is the number of losses. The two quantities must sum up to the number of gambles, $n_1 + n_2 = T$, which in our case is $T = 100$. Now the amount of money a solitary player is expected to have tomorrow can be obtained by first extending the time horizon

$$\lim_{T \rightarrow \infty} (1.5^{n_1} * 0.6^{n_2})^{\frac{1}{T}}$$

Due to the law of large numbers, quantities n_1 and n_2 will both converge to $\frac{T}{2}$.

$$\lim_{T \rightarrow \infty} (1.5^{\frac{T}{2}} * 0.6^{\frac{T}{2}})^{\frac{1}{T}}$$

$$\lim_{T \rightarrow \infty} (1.5 * 0.6)^{(\frac{T}{2} \frac{1}{T})}$$

$$\lim_{T \rightarrow \infty} (1.5 * 0.6)^{\frac{1}{2}} = 0.95$$

This is the long term average per period that the wealth is expected to change. The amount of money our gambler is expected to have after 100 days is then

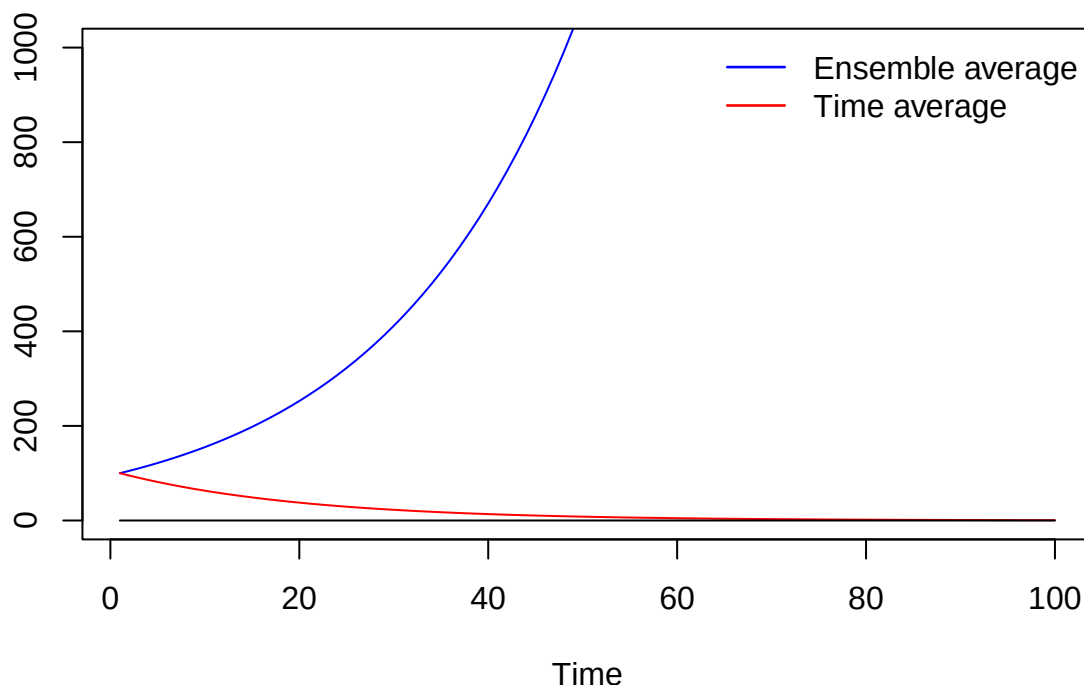
$$0.95^{100} * 100 = 0.6\$$$

By the end of the experiment, the individual gambler is expected to be below the one dollar entrance requirement, hence, bust. The process describing an independent gambler is non-ergodic, with

$$\bar{W}(t) \neq \bar{W}(i)$$

The figure illustrates how the two expected values diverge from each other over the 100 period window.

```
x <- c(100)
y <- x
for(i in 2:100) {
  x[i] <- x[i-1]*1.05
  y[i] <- y[i-1]*0.95
}
ts.plot(ts(x), ts(y), 0, col=c("blue", "red", "black"), ylim=c(0,1000))
legend("topright", legend=c("Ensemble average", "Time average"),
      col=c("blue", "red"), lwd=1.3, bty="n")
```



Recall the definition of ergodicity from the beginning. The first case is ergodic due to the redistribution scheme: the wealth of one player grows at the same pace as the average wealth in the group. However, in the second case we saw that the time average for an individual is not the same as the ensemble average. The multiplicative dynamics and path dependency makes the game non-ergodic.

The key takeaway: With non-ergodic phenomena the ensemble average over different realizations does not describe the typical time evolution of a single realization.

Simulation

This following R-code simulates the case 2 of the above gamble for N independent players (no pooling). This setup was proposed in Peters (2018). The idea is to demonstrate how differently ensemble averages and time averages can function in a simple non-ergodic settings. Peters is quite polemical in the writing and he has been critiqued for overstating the scope of the problem in economic models.

Conclusion: With N and T tending to infinity, the share of gamblers going bust approaches 100%, but at the same time the highest winnings grow even faster. How can the ensemble average grow if the expected time average of each player shrinks? Because some of the players (measure zero in the limit) are so lucky that their rapidly increasing winnings compensate for the gradually declining losses of the rest.

```
gamble <- function(type, rounds, players, payoffs, cutoff){

  stoppages <- c()

  X <- matrix(0, nrow=rounds, ncol=players)

  for(j in 1:players) {
    x <- c(100)
    stoppages[j] <- rounds

    for(i in 2:rounds) {
      if(type=="additive") {
        ifelse(sample(0:1,1)==0,
              change <- payoffs[1],
              change <- payoffs[2])
      } else {
        ifelse(sample(0:1,1)==0,
              change <- payoffs[1] * x[i-1],
              change <- payoffs[2] * x[i-1])
      }

      x[i] <- x[i-1] + change

      if(x[i] <= cutoff) {
        stoppages[j] <- i
        break
      }
    }

    X[1:length(x),j] <- x
  }

  mean_run <- apply(X, 1, mean) # Ensemble average of players at each round
  mean_stoppage <- round(mean(stoppages) ,0)
  survivors <- length(which(stoppages == rounds))/players*100

  ts.plot(ts(X), ts(mean_run), line=100, line=0,
         col=c(rep("forestgreen", players), "red", "black", "black"),
         xlim=c(0, rounds), lwd=c(rep(0.3, players), 2, 1, 1), ylab="Value")
}
```

```

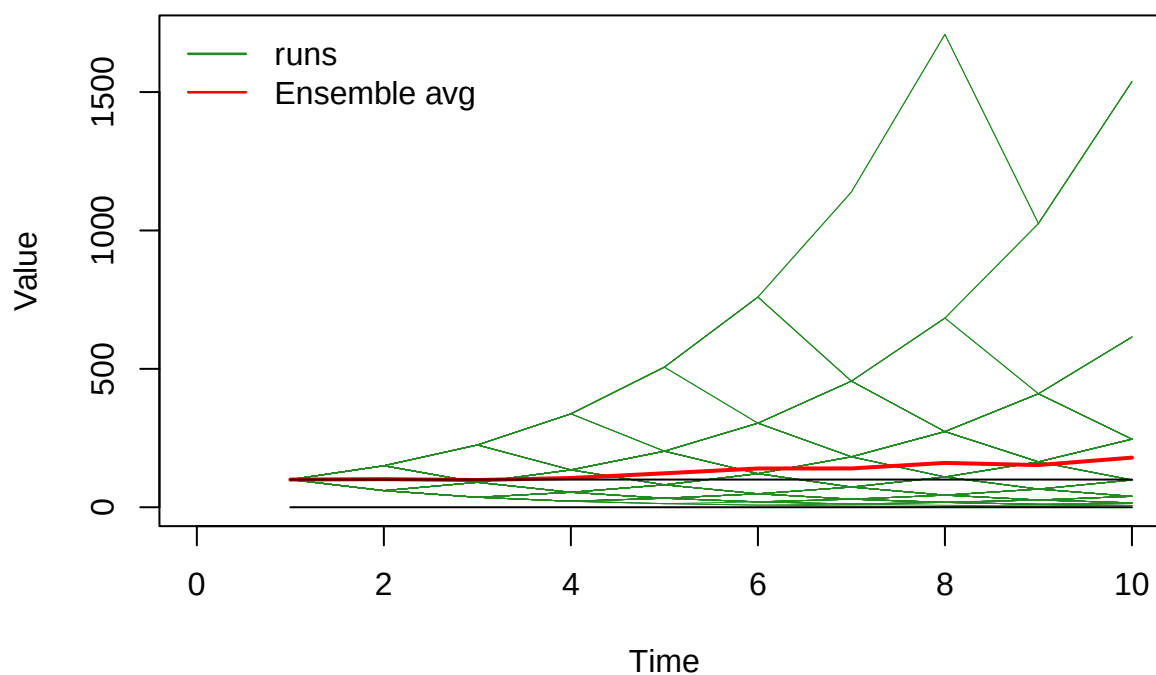
legend("topleft", legend=c("runs", "Ensemble avg"),
      col=c("forestgreen", "red"), bty="n", lwd=1.3)
if(type=="additive") {
  title(paste("Additive gamble with ", paste(players),
    " players and ", paste(rounds), " periods"))
} else {
  title(paste("Multiplicative gamble with ", paste(players),
    " players and ", paste(rounds), " periods"))
}

if(players==1) {
  cat("\n Game stopped after: ", mean_stoppage, "/",
    rounds, "\n Highest value: ", max(X))
} else {
  cat("\n Average player survived until round: ", mean_stoppage, "/",
    rounds, " rounds",
    "\n Players that made it to the end: ", survivors, "%",
    "\n Highest winnings in the end: ", round(max(X[nrow(X)]),0), "$",
    "\n Ensemble average in the end: ", round(mean_run[length(mean_run)],0), "$")
}
}

gamble(type="multiplicative", # "multiplicative/additive"
  rounds=10,                 # max tosses per player
  players=100,               # Number of gamblers
  payoffs=c(0.5,-0.4),       # c(win,lose) either multipliers or actual sums
  cutoff=1                   # definition of ruin
)

```

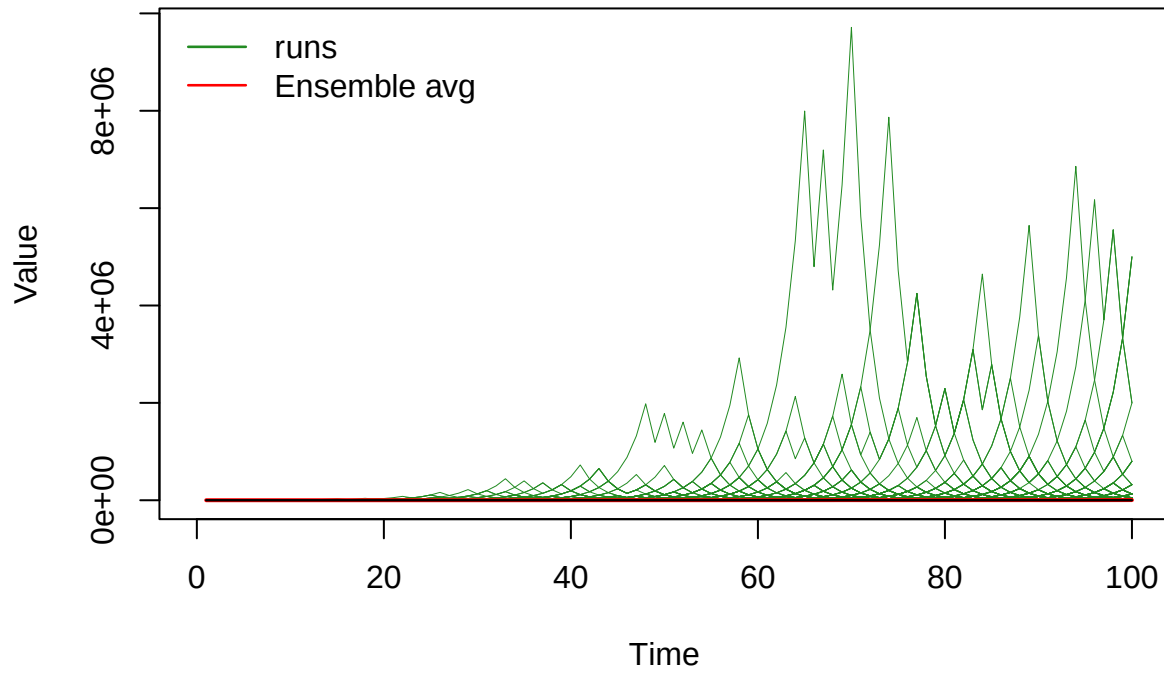
Multiplicative gamble with 100 players and 10 periods



```
##
## Average player survived until round: 10 / 10 rounds
## Players that made it to the end: 100 %
## Highest winnings in the end: 1538 $
## Ensemble average in the end: 179 $
```

```
gamble(type="multiplicative",
  rounds=100,
  players=10000,
  payoffs=c(0.5,-0.4),
  cutoff=1
)
```

Multiplicative gamble with 10000 players and 100 periods



```
##  
## Average player survived until round: 66 / 100 rounds  
## Players that made it to the end: 31.18 %  
## Highest winnings in the end: 4999820 $  
## Ensemble average in the end: 3476 $
```